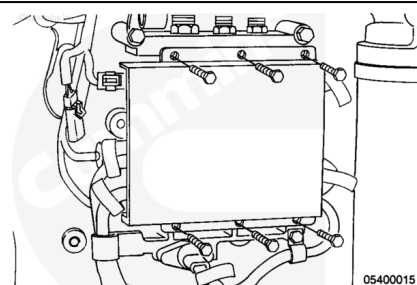


Trainings ▾

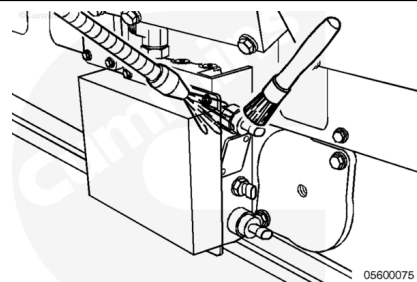
Remove with Mechanically Actuated Injector

Remove the ECM. Refer to Procedure 019-031 (</qs3/pubsys2/xml/en/procedures/19/19-019-031.html>) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.



⚠ WARNING ⚠

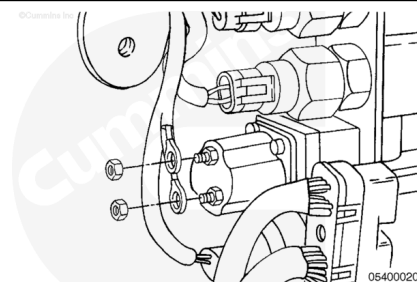
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the fuel shutoff valve and surrounding area.

Use a 3/8-inch wrench to remove the nut holding the electrical connection of the fuel shutoff valve coil.

Remove the connection.



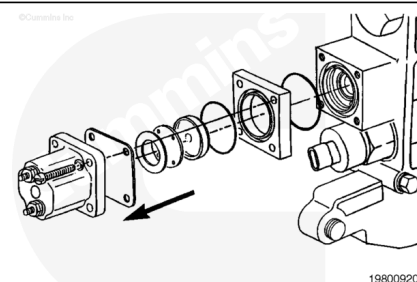
Remove the four capscrews.

Remove the coil housing and the fuel shield.

Discard the o-ring.

Remove the spring washer, valve disc, actuator disc, and actuator spacer from the valve housing.

Discard the o-ring.



Disassemble

with Mechanically Actuated Injector

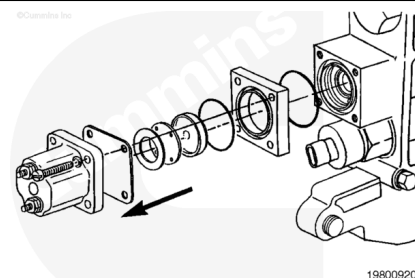
Remove the four capscrews.

Remove the coil housing and the fuel shield.

Discard the o-ring.

Remove the spring washer, valve disc, actuator disc, and actuator spacer from the valve housing.

Discard the o-ring.



Clean and Inspect for Reuse

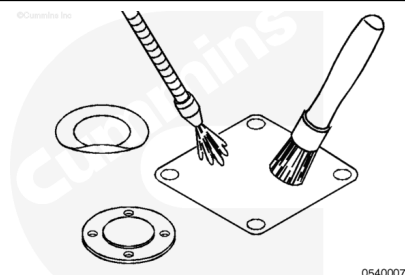
with Mechanically Actuated Injector

⚠ WARNING ⚠

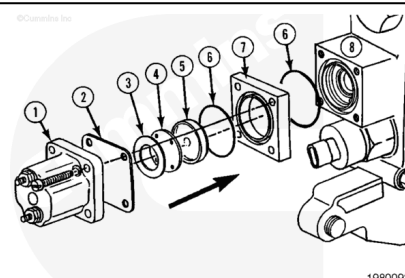
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Use mineral spirits to clean all of the parts except the coil assembly.

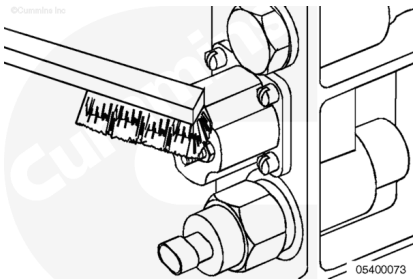
Do **not** get solvent on the coil. Clean the coil with a dry cloth.
Use a 200-grit emery cloth and a flat surface to polish the coil surface.



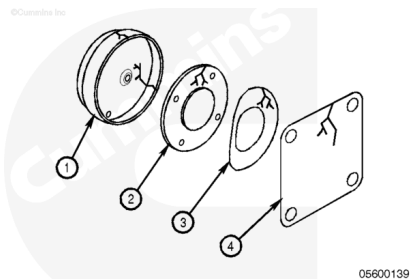
Inspect the fuel shut off valve housings (7) and (8) for wear and other damage.



Use a wire brush to clean any corrosion from the coil terminal.



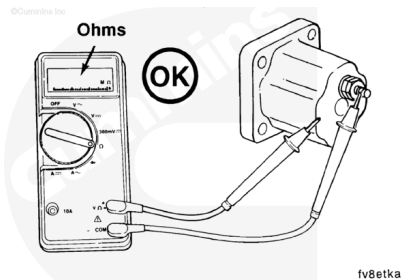
Inspect the valve disc (2), valve spring (3), actuator disc (1), and shutoff valve shield (4) for dirt, metal parts, bonding separation, corrosion, cracks, or wear. Replace if necessary.



Measure the resistance of the coil with a multimeter.

Fuel System Shutoff Valve Specifications

Voltage	Resistance Minimum (Ohms)	Resistance Maximum (Ohms)
6 VDC	1	5
12 VDC	6	15
24 VDC	24	50
32 VDC	42	80
36 VDC	46	87
48 VDC	92	145
74 VDC	315	375
115 VAC	645	735



Replace the coil if it does **not** meet the specifications.

If the coil assembly shows 0 ohms, there is an electrical short in the coil.

Assemble

with Mechanically Actuated Injector

Install a new o-ring (6) into the electronic control valve body (8).

Install the actuator housing (7) with the o-ring groove toward the coil (1).

Install the actuator disc (5) with the cup side toward the coil (1).

Place the rubber side to the valve disc (4) on the actuator disc (5).

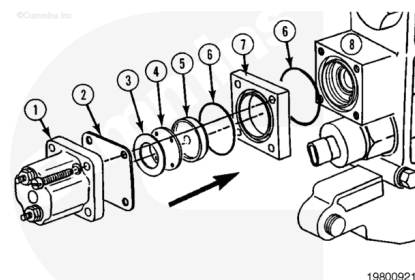
Install the valve spring (3) with the cup side toward the coil. The inner diameter of the spring (3) **must** rest on the pilot diameter of the valve disc (4).

Install a new o-ring (6) into the actuator housing (7).

Install the fuel shield (2) and coil (1) on to the front cover (8).

Tighten the four capscrews.

Torque Value: 8 n•m [71 in-lb]



Install

with Mechanically Actuated Injector

Install a new o-ring (6) into the electronic control valve body (8).

Install the actuator housing (7) with the o-ring groove toward the coil (1).

Install the actuator disc (5) with the cup side toward the coil (1).

Place the rubber side to the valve disc (4) on the actuator disc (5).

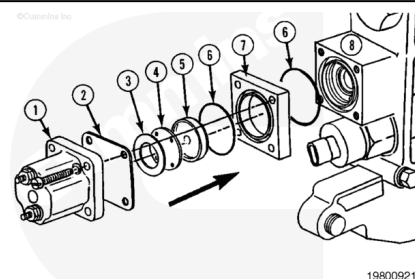
Install the valve spring (3) with the cup side toward the coil. The inner diameter of the spring (3) **must** rest on the pilot diameter of the valve disc (4).

Install a new o-ring (6) into the actuator housing (7).

Install the fuel shield (2) and coil (1) on to the front cover (8).

Tighten the four capscrews.

Torque Value: 8 n•m [71 in-lb]

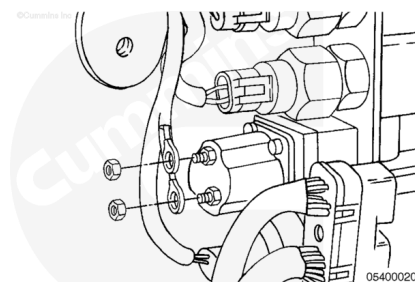


Install the electrical connection on the fuel shutoff valve.

Install the nut on the threaded post of the coil.

Using two 3/8-inch wrenches, hold the post of the nut firmly while tightening the connection nut.

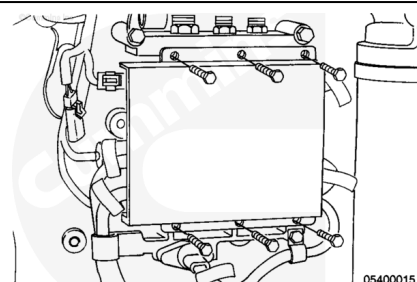
Torque Value: 2 n•m [18 in-lb]



Install the ECM to the control valve body. Refer to Procedure 019-031 (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.

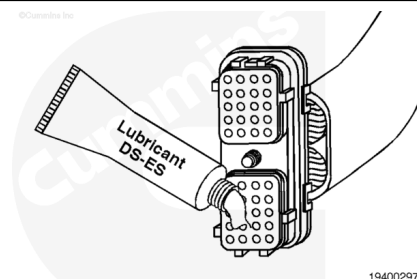
Tighten the six ECM mounting capscrews.

Torque Value: 8 n•m [71 in-lb]



⚠ CAUTION ⚠

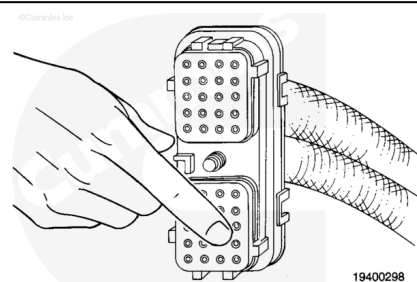
Use only Cummins recommended lubricant DS-ES. Other lubricating oil or grease in the connectors can cause ECM damage, poor engine performance, or premature connector pin wear.



Apply a thin coat of lubricant to the connector nosepiece.

Spread the lubricant across the connector nosepiece so it enters every pin hole and lubricates the contacts.

Lubricant **must not** be visible on the surface of the nosepiece.



Connect the OEM and engine harness connectors to the ECM.

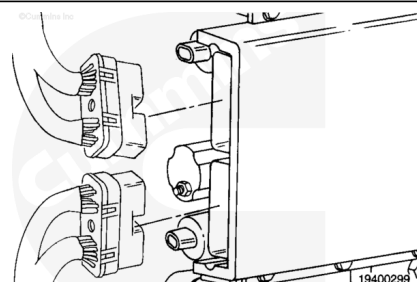
Tighten the connector capscrews.

Torque Value: 3 n•m [27 in-lb]

When an ECM is replaced, the new ECM **must** be calibrated.

Refer to Procedure 019-032

(/qs3/pubsys2/xml/en/procedures/105/105-019-032.html) in



the Troubleshooting and Repair, Electronic Control System,
QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin
3666113.

Last Modified: 23-Dec-2005
